

Vertical Pod Autoscaler (VPA)

- **Vertical Pod Autoscaler (VPA)** is a CRD that **adjusts resource allocations** to individual containers based on their usage patterns in order to optimize resource usage and reduce wastage
- VPA scales the resources allocated to individual pods vertically, in contrast to the Horizontal Pod Autoscaler (HPA), which scales the number of replicas horizontally
- VPA can be used to **optimize CPU, memory**, or other specific resources for containers
- VPA operates in four different modes:
 - **Auto**: VPA allocates resource requests and modifies them for existing pods as well as newly created pods. Does it once restart free updates are available
 - **Recreate**: VPA allocates resource requests and modifies them for existing pods as well as newly created pods. It evicts them when the requested resources significantly differ from the new recommendation
 - **Initial**: only allocates resource requests to newly created pods and never modifies them afterward
 - **Off**: VPA does not automatically alter the resource requests of pods but still provides suggestions that can be viewed
- VPA has three components:
 - **Recommender**: makes resource recommendations for pods
 - **Updater** : applies VPA recommendations automatically without manual intervention
 - **Admission plugin**: sets the appropriate resource requests